# Summer 2024
# ROUND 2
## August 9-12, 2024

# Instructions and Information

1. This is a 7-question, 3-day test, in which all questions require proofs. Each question has a given weight (number of points) shown in the test beside the problem. The maximum number of points you can get is 50.
2. You may quote well-known facts without proof. Otherwise, please provide the proof of the fact. If you are unsure if a fact is well-known enough, please ask stayhomedomath.
3. For all questions, please submit your solutions as separate PDFs to your assigned Discord channel and pin the files there.
4. No calculators, books, graphing software, online resources, etc. are allowed. However, we grant permission to use word processing applications (collaborative or not), e.g. Google Docs, Overleaf, etc. so you may collaborate with your team. For physical supplies, only blank scratch paper, ruler, compass, and writing materials are allowed. If anyone is caught with illegal resources, they will be disqualified.
5. All communications must be done in the team channels in the Mathematics server.
6. No person who participated or helped in the contest is allowed to discuss any aspect of it before the round ends. If moderators catch anyone discussing during this time frame, they will automatically be disqualified.
7. Ties among two teams will be broken according to the team who scores higher in the last problem that their scores differ, then further ties will be broken according to their score in Round 1.

# §1 Problems

1. **[6]** Let $f : \mathbb{R}^+ \to \mathbb{R}^+$ be a function such that
$$f(x)f(y) = f(xy) + f\left(\frac{x}{y}\right)$$
for $x, y \in \mathbb{R}^+$. Prove that $f(x) \geq 2$ for all $x \in \mathbb{R}^+$.

2. **[6]** Let $k$ be a positive integer. Given a set $S$ containing all positive integers at most $2k$, find the minimum positive integer $n \geq 2$, such that for any subset $X$ of $S$ that has size $n$, $X$ always has distinct elements $a, b$ such that $a + b$ is a prime.

3. **[6]** Call a real number $x$ "repetitive" if for every positive integer $k$ there exists a string of $k$ digits that appears more than once in the decimal expansion of $x$. (Numbers with terminating decimals are written with infinitely many 0s at the end.) Prove that if $x$ is repetitive, then $x^2$ is repetitive.

4. **[7]** In $\triangle ABC$, $AB = AC$. $E$ is a point on $BC$ such that $AE = BE$. $G$ is a point opposite to $B$ with respect to side $AC$ such that $\triangle AGC \cong \triangle AEB$. Let $H$ be the intersection of $EG$ and $AC$, and let $I$ be the intersection of $BH$ and $AE$. Prove that the intersection of $BG$ and $CI$ lies on the circumcircle of $\triangle ACE$.

5. **[8]** Culver is on the number line, standing at 0, where the positive direction of the number line is to his right. He randomly chooses a string consisting of exactly 2024 of each of `L` and `R`. He also has a score that is initially 1.
At the $i$-th second for $1 \leq i \leq 4048$,
   - If the $i$-th character in the string is `L`, Culver multiplies his score by the position he is currently at, then moves left 1 unit;
   - Otherwise, he moves right 1 unit.

   What is his expected final score?

6. **[9]** Let $r > 1$ be a real constant. Find the smallest constant $C$ in terms of $r$, such that for all positive integers $n$ and real numbers $1 = a_0 \leq a_1 \leq \ldots \leq a_n = r$, we have
$$\sum_{i=1}^{n} \left( a_i^2 a_{i-1} - a_{i-1}^2 a_i + \frac{a_i^2}{a_{i-1}} - \frac{a_{i-1}^2}{a_i} \right) \leq C$$

7. **[9]** Let $n$ be a positive integer. Culver has a $(n+1) \times (n+1)$ grid, and he wants to colour each of its cells with a colour such that
   - The first row is same as the last row.
   - The first column is same as the last column.
   - Each $2 \times 2$ subgrid has a distinct colour pattern.

   At least how many distinct colours does he have to use?